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Sutherland's law

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In 1893 William Sutherland ([http://en.wikipedia.org/wiki/William_Sutherland_\(physicist\)](http://en.wikipedia.org/wiki/William_Sutherland_(physicist))) , an Australian physicist, published a relationship between the dynamic viscosity, μ , and the absolute temperature, T , of an ideal gas. This formula, often called Sutherland's law, is based on kinetic theory of ideal gases and an idealized intermolecular-force potential. Sutherland's law is still commonly used and most often gives fairly accurate results with an error less than a few percent over a wide range of temperatures. Sutherland's law can be expressed as:

$$\mu = \mu_{ref} \left(\frac{T}{T_{ref}} \right)^{3/2} \frac{T_{ref} + S}{T + S}$$

T_{ref} is a reference temperature.

μ_{ref} is the viscosity at the T_{ref} reference temperature

S is the Sutherland temperature

Some authors instead express Sutherland's law in the following form:

$$\mu = \frac{C_1 T^{3/2}}{T + S}$$

Comparing the formulas above the C_1 constant can be written as:

$$C_1 = \frac{\mu_{ref}}{T_{ref}^{3/2}} (T_{ref} + S)$$

Sutherland's law coefficients:

Gas	$\mu_0 [\frac{kg}{ms}]$	$T_0 [K]$	$S [K]$	$C_1 [\frac{kg}{ms\sqrt{K}}]$
Air	1.716×10^{-5}	273.15	110.4	1.458×10^{-6}

References

- **Sutherland, W. (1893)**, "The viscosity of gases and molecular force", Philosophical Magazine, S. 5, 36, pp. 507-531 (1893).

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